

```

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

```

```

# Load the Wine dataset
wine = load_wine()

# Features (independent variables)
X = wine.data

# Target labels (dependent variable)
y = wine.target

```

```

print("Dataset Description:\n")
print(wine.DESCR)

print("\nShape of Dataset:", X.shape)
print("Number of Samples:", X.shape[0])
print("Number of Features:", X.shape[1])

print("\nFeature Names:")
print(wine.feature_names)

print("\nTarget Classes:")
print(wine.target_names)

```

Dataset Description:

.. _wine_dataset:

Wine recognition dataset

****Data Set Characteristics:****

:Number of Instances: 178

:Number of Attributes: 13 numeric, predictive attributes and the class

:Attribute Information:

- Alcohol
- Malic acid
- Ash
- Alcalinity of ash
- Magnesium
- Total phenols
- Flavanoids
- Nonflavanoid phenols
- Proanthocyanins
- Color intensity
- Hue
- OD280/OD315 of diluted wines
- Proline
- class:
 - class_0
 - class_1
 - class_2

:Summary Statistics:

```
=====
              Min    Max    Mean    SD
=====
Alcohol:      11.0  14.8   13.0   0.8
Malic Acid:   0.74  5.80   2.34  1.12
Ash:          1.36  3.23   2.36  0.27
Alcalinity of Ash: 10.6 30.0   19.5  3.3
Magnesium:    70.0 162.0  99.7  14.3
Total Phenols: 0.98  3.88   2.29  0.63
Flavanoids:   0.34  5.08   2.03  1.00
Nonflavanoid Phenols: 0.13 0.66   0.36  0.12
Proanthocyanins: 0.41  3.58   1.59  0.57
Colour Intensity: 1.3  13.0   5.1   2.3
Hue:          0.48  1.71   0.96  0.23
OD280/OD315 of diluted wines: 1.27 4.00   2.61  0.71
Proline:      278  1680   746   315
=====
```

:Missing Attribute Values: None
 :Class Distribution: class_0 (59), class_1 (71), class_2 (48)
 :Creator: R.A. Fisher
 :Donor: Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)
 :Date: July, 1988

This is a copy of UCI ML Wine recognition datasets.

```
df = pd.DataFrame(X, columns=wine.feature_names)
df["Class"] = y

df.head()
```

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins	col
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	0.28	2.29	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	0.26	1.28	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	0.30	2.81	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	0.24	2.18	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	0.39	1.82	

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   alcohol                               178 non-null    float64
1   malic_acid                            178 non-null    float64
2   ash                                   178 non-null    float64
3   alcalinity_of_ash                     178 non-null    float64
4   magnesium                             178 non-null    float64
5   total_phenols                         178 non-null    float64
6   flavanoids                            178 non-null    float64
7   nonflavanoid_phenols                  178 non-null    float64
8   proanthocyanins                       178 non-null    float64
9   color_intensity                       178 non-null    float64
10  hue                                   178 non-null    float64
11  od280/od315_of_diluted_wines          178 non-null    float64
12  proline                               178 non-null    float64
13  Class                                 178 non-null    int64
dtypes: float64(13), int64(1)
memory usage: 19.6 KB
```

df.describe()

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols	proanth
count	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000	178.000000
mean	13.000618	2.336348	2.366517	19.494944	99.741573	2.295112	2.029270	0.361854	0.361854
std	0.811827	1.117146	0.274344	3.339564	14.282484	0.625851	0.998859	0.124453	0.124453
min	11.030000	0.740000	1.360000	10.600000	70.000000	0.980000	0.340000	0.130000	0.130000
25%	12.362500	1.602500	2.210000	17.200000	88.000000	1.742500	1.205000	0.270000	0.270000
50%	13.050000	1.865000	2.360000	19.500000	98.000000	2.355000	2.135000	0.340000	0.340000
75%	13.677500	3.082500	2.557500	21.500000	107.000000	2.800000	2.875000	0.437500	0.437500
max	14.830000	5.800000	3.230000	30.000000	162.000000	3.880000	5.080000	0.660000	0.660000

```
df.isnull().sum()
```

	0
alcohol	0
malic_acid	0
ash	0
alcalinity_of_ash	0
magnesium	0
total_phenols	0
flavanoids	0
nonflavanoid_phenols	0
proanthocyanins	0
color_intensity	0
hue	0
od280/od315_of_diluted_wines	0
proline	0
Class	0

```
dtype: int64
```

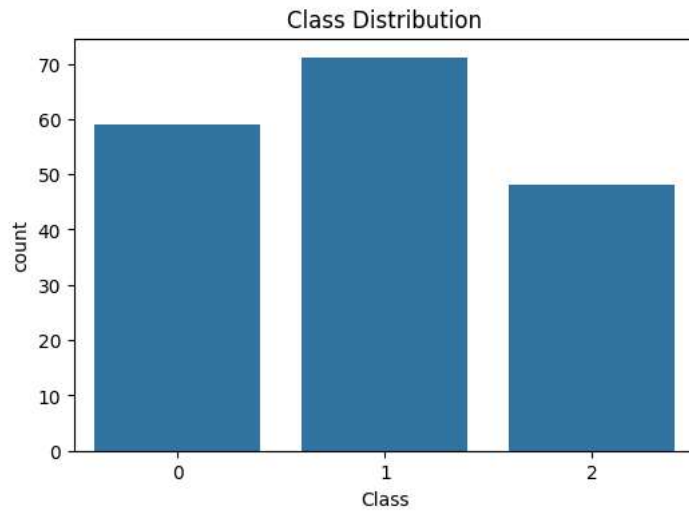
```
print("Duplicate Rows:", df.duplicated().sum())
```

```
Duplicate Rows: 0
```

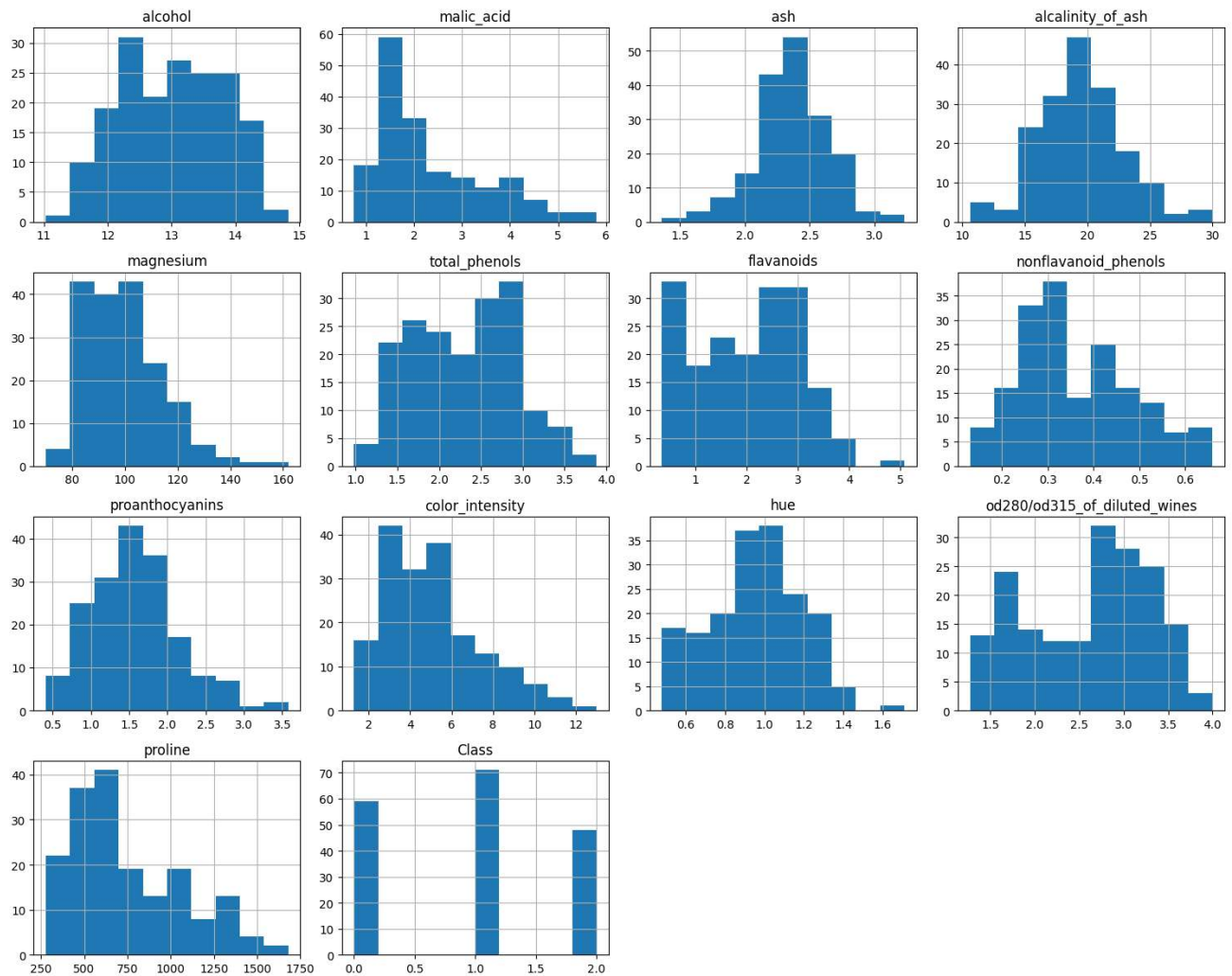
```
print(df["Class"].value_counts())

plt.figure(figsize=(6,4))
sns.countplot(x="Class", data=df)
plt.title("Class Distribution")
plt.show()
```

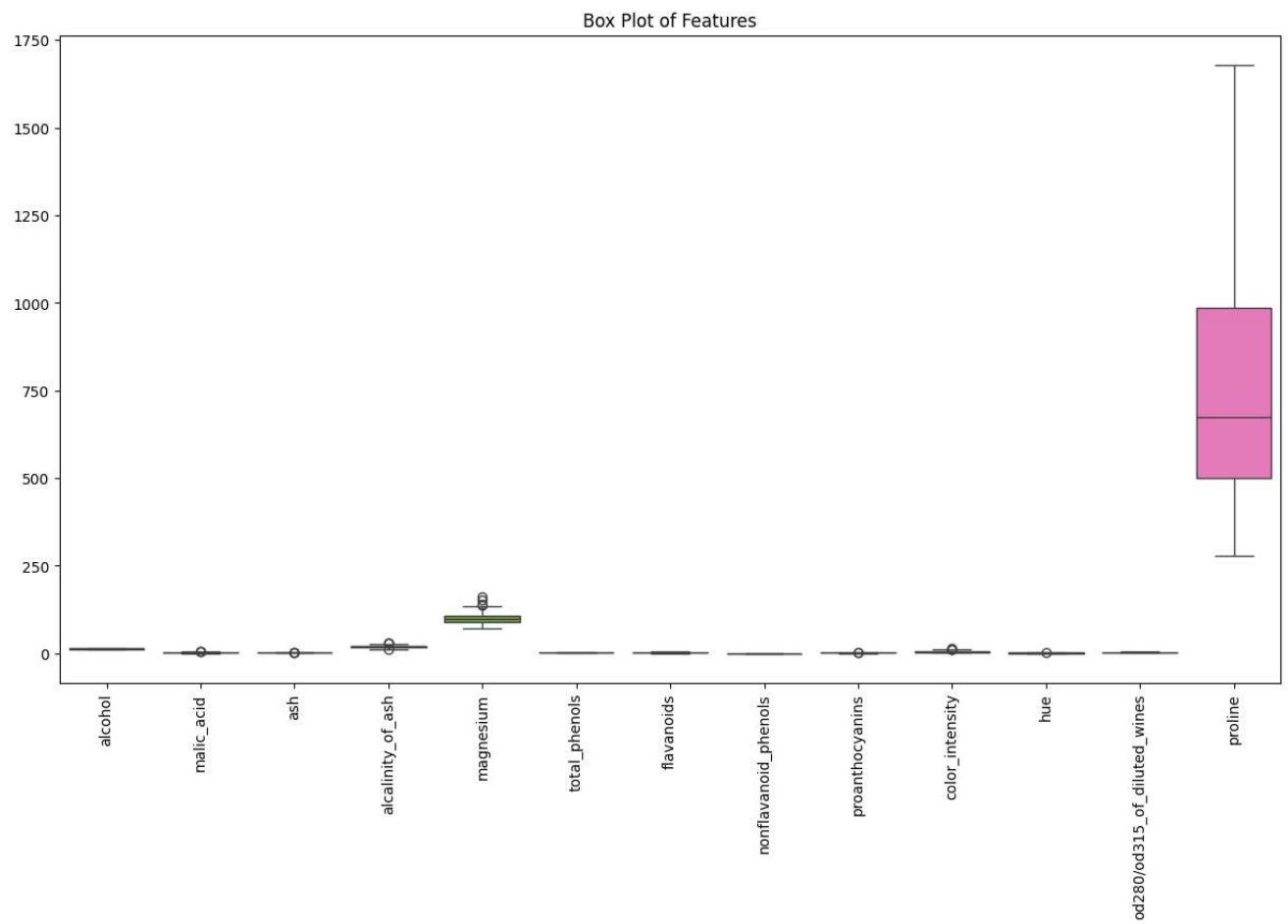
```
Class
1    71
0    59
2    48
Name: count, dtype: int64
```



```
df.hist(figsize=(15,12))
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(15,8))
sns.boxplot(data=df.iloc[:, :13])
plt.xticks(rotation=90)
plt.title("Box Plot of Features")
plt.show()
```



```
sns.pairplot(df, hue="Class", diag_kind="hist")  
plt.show()
```

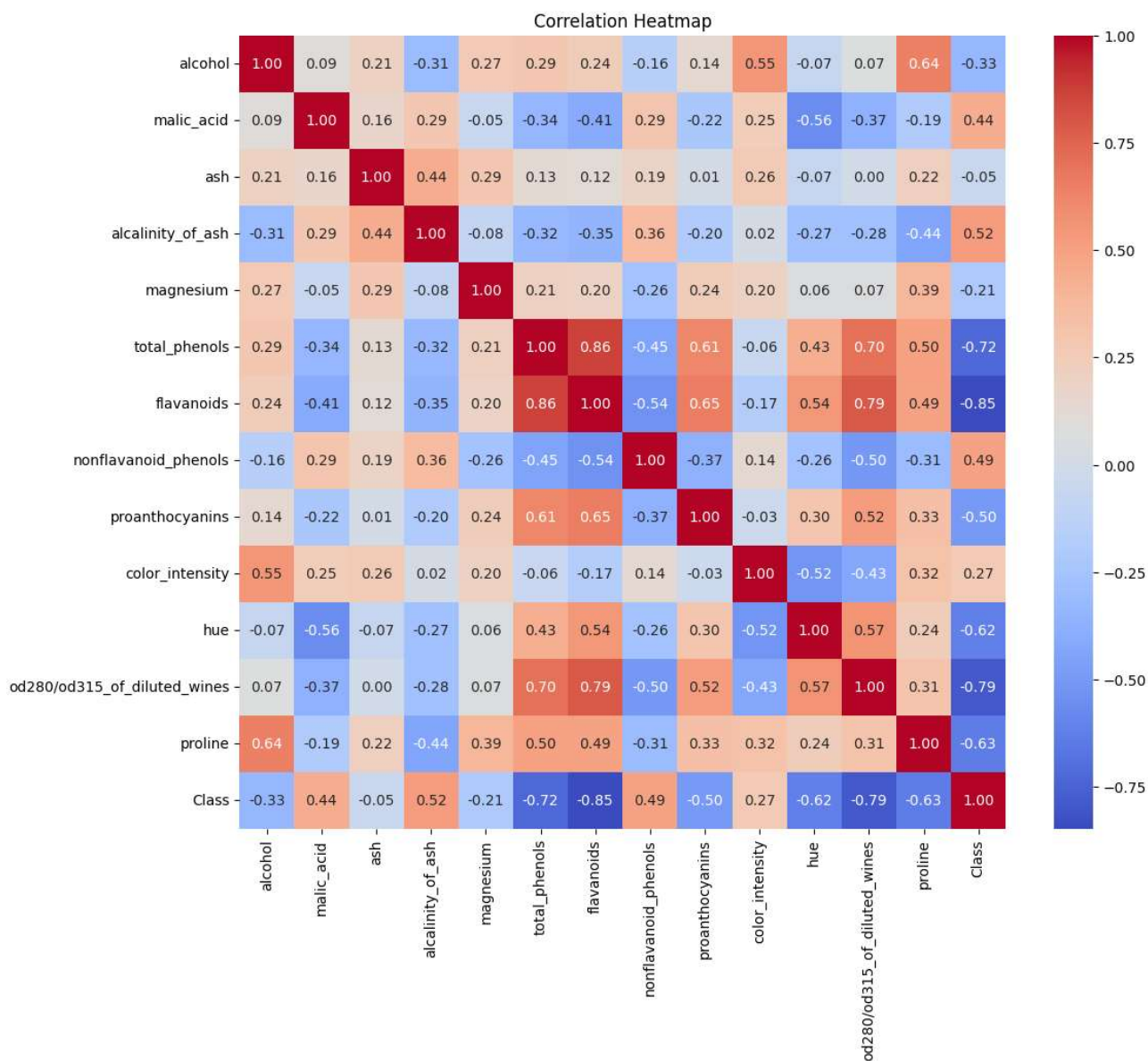


```
plt.figure(figsize=(12,10))
```

```
sns.heatmap(
    df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
```



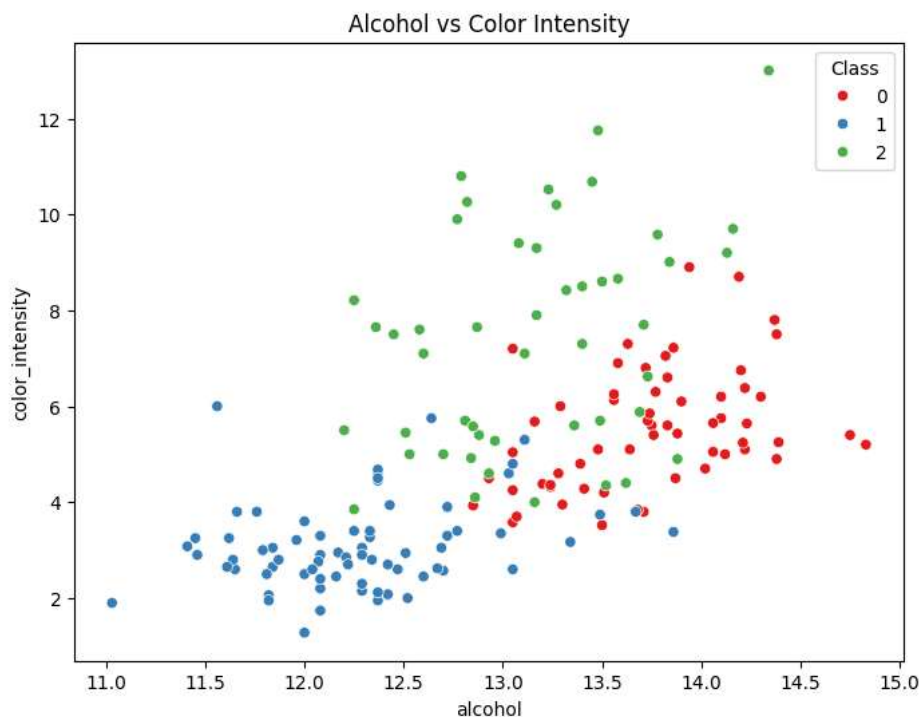
```
plt.title("Correlation Heatmap")
plt.show()
```



```
plt.figure(figsize=(8,6))

sns.scatterplot(
    data=df,
    x="alcohol",
    y="color_intensity",
    hue="Class",
    palette="Set1"
)

plt.title("Alcohol vs Color Intensity")
plt.show()
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)
```

```
mlp = MLPClassifier(
    hidden_layer_sizes=(10,10),
    activation="relu",
    solver="adam",
    max_iter=1000,
    random_state=42
)
```

```
mlp.fit(X_train, y_train)
```

```
MLPClassifier
MLPClassifier(hidden_layer_sizes=(10, 10), max_iter=1000, random_state=42)
```

```
plt.figure(figsize=(8,5))

plt.plot(mlp.loss_curve_, linewidth=2)

plt.title("Training Loss Curve")
plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.grid(True)

plt.show()
```



```
y_pred = mlp.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", round(accuracy*100,2),"%")
```

```
Accuracy: 97.22 %
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
plt.figure(figsize=(6,5))
```

```
sns.heatmap(  
    cm,  
    annot=True,  
    fmt="d",  
    cmap="Blues",  
    xticklabels=wine.target_names,  
    yticklabels=wine.target_names  
)
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.title("Confusion Matrix")
```